

MEVAN WIJewardena

+1(213) 649-8651 · <https://mevanwijewardena.github.io/> · mpathira@usc.edu · www.linkedin.com/in/mevan96

RESEARCH OVERVIEW

My research develops learning and control algorithms for stochastic systems with partial feedback, focusing on coupling across agents, resources, and time. I combine tools from stochastic optimization, bandit learning, and game theory to design provably efficient and practically relevant methods.

EDUCATION

University of Southern California, Los Angeles, USA

Aug. 2022 - Present

Ph.D. in Electrical Engineering

CGPA 4.0 out of 4.0

Advisor - Prof. Michael James Neely

University of Moratuwa, Moratuwa, Sri Lanka

Oct. 2016 - July 2021

B.Sc. Engineering Honours Degree specializing in Electronic and Telecommunication Engineering

First Class Honors with a CGPA 4.06 out of 4.2 (Ranked **5** out of **101**)

Dean's List Placements - Semesters **1, 2, 3, 4, 7, 8**

PUBLICATIONS

- **M. Wijewardena**, and M. J. Neely, “Online Multi-Player Resource-Sharing Games with Bandit Feedback,” Dynamic Games and Applications. 2025. [<https://tinyurl.com/2yzaykxw>]
- **M. Wijewardena**, and M. J. Neely, “Adaptive Algorithms for Automatic Link Selection in Multiple Access with Link Failures,” in 23rd International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (WiOpt '25), May 2025. [<https://ieeexplore.ieee.org/document/11123300>]
- **M. Wijewardena**, M. J. Neely, “A Two-Player Resource-Sharing Game with Asymmetric Information,” Games. 2023; 14(5):61. [<https://doi.org/10.3390/g14050061>]
- **M. Wijewardena**, T. Samarasinghe, K. T. Hemachandra, S. Atapattu and J. S. Evans, “Physical Layer Security for Intelligent Reflecting Surface Assisted Two-Way Communications,” in IEEE Communications Letters, vol. 25, no. 7, pp. 2156-2160, July 2021. [<https://ieeexplore.ieee.org/document/9383283>]
- R. Boldu, **M. Wijewardena**, H. Zhang, and S. Nanayakkara, “MAGHair: A Wearable System to Create Unique Tactile Feedback by Stimulating Only the Body Hair,” in 22nd International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '20), Oct. 2020. [<https://dl.acm.org/doi/10.1145/3379503.3403545>]
- **M. Wijewardena**, K. Asgari, and M. J. Neely, “Bandit-Based Rate Adaptation for a Single-Server Queue,” under consideration at Stochastic Systems.
- **M. Wijewardena**, M. J. Neely, and Haipeng Luo, “Automatic Link Selection in Multi-Channel Multiple Access with Link Failures,” under consideration at IEEE Transactions on Networking.

RESEARCH TALKS

- WiOpt '25 - Linköping, Sweden. May 2025
 - The 23rd International Symposium on Modeling and Optimization in Mobile, Ad Hoc, and Wireless Networks
 - Topic: Adaptive Algorithms for Automatic Link Selection in Multiple Access with Link Failures.
- University of Moratuwa - ENTC Research Seminar Aug. 2021
 - The monthly research seminar organized by the Department of Electronic and Telecommunication Engineering, University of Moratuwa.
 - Topic: Physical Layer Security for Intelligent Reflecting Surface Assisted Two-Way Communications.

CONFERENCE/JOURNAL REVIEWER

- IEEE/ACM Transactions on Networking
- Springer Cluster Computing
- 2025 Moratuwa Engineering Research Conference (MERCon)
- 2021 IEEE EMBS International Student Conference (Reviewer ID - R015)

WORK EXPERIENCE

University of Southern California, Los Angeles, USA Aug. 2023 - Present

Teaching Assistant

Department of Electrical and Computer Engineering

- Teaching Assistant for multiple courses in stochastic processes, optimization, and networks (EE 512, EE 518, EE 450, EE 538)
- Led discussion sessions, prepared problem sets and HWs.

University of Moratuwa, Moratuwa, Sri Lanka

June 2021 - July 2022

Lecturer (On Contract)

Department of Electronic and Telecommunication Engineering

- Teaching Assistant for multiple courses in Communications, Signal Processing, Networks, Electronics and Computer Organization (EN2073, EN2040, EN2053, EN2090, EN2022, EN1094)
- Conducted laboratory sessions, prepared problem sets and HWs.

University of Auckland, New Zealand

June 2019 - Dec. 2019

Visiting Researcher

Augmented Human Lab

- Worked on two research projects (MAGHair and AiSee) in Human Computer Interaction

SELECTED ACADEMIC ACHIEVEMENTS

- International Mathematical Olympiad (IMO) — **Bronze Medal** ×4 2012, 2013, 2014, 2016
- Asia Pacific Mathematical Olympiad (APMO) — **Silver Medal** 2016
- International Mathematics Competition for University Students — **Bronze Medal** 2018
- IEEEExtreme Programming Competition — World Rank **52**, Country Rank **1** 2020
- Google Code Jam — World Rank **1011**, Country Rank **2** 2019
- Sri Lanka Mathematical Olympiad — **Gold Medal** ×4 2012–2016

RESEARCH CONTRIBUTIONS

NUM with Partial Feedback, and Restless Markovian Channels

June 2026 -Present

- Developing algorithms for maximizing network utility under bandit feedback.

- Deviating from the classical i.i.d. assumptions, we consider a setting where channels evolve as restless Markov chains with unknown channel statistics.
- Develops one of the first learning frameworks integrating NUM, bandit feedback and Markovian channels.

Bandit-Based Rate Adaptation for a Single-Server Queue

Jan. 2025 - June 2026

- Developed provably efficient learning algorithms for queue stability under bandit feedback with continuous action spaces; established lower bounds on worst-case time-average queue size.
- Develops one of the first learning frameworks integrating queueing theory with continuum armed bandits.

Automatic Link Selection in Multiple Access with Link Failures

Apr. 2024 - May 2026

- Designed provably convergent learning algorithms for resource allocation under stochastic link failures with matching constraints.
- This work combines network utility maximization with combinatorial bandits, in non stationary environments.

Multi-Player Resource-Sharing Games with Fair Reward Allocation

July 2023 - Feb. 2024

- Developed algorithms for worst-case utility maximization in multi-agent systems with bandit feedback and coupling constraints.

Secrecy of two-way IRS communication systems

Feb. 2020 - July 2021

- Developed algorithms for the sum-secrecy rate maximization in a two-way Intelligent Reflecting Surfaces (IRS) aided communication system in the presence of an eavesdropper.
- This work was the first to jointly address, two-way IRS systems, and secrecy.

OTHER PROJECTS

Safety of Vision Language Models

April 2026 - Present

- Investigating the mechanisms by which vision-language models detect unsafe multimodal inputs, where the image and text are individually benign but jointly constitute unsafe content.
- Conducting causal intervention experiments on Qwen and LLaVA models to identify the components responsible for multimodal safety behavior and quantify their contribution to multimodal safety decisions.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, MATLAB
- **Machine Learning:** Bandit Learning, Online Learning, Statistical Learning, Time Series Analysis, Attention-based Neural Networks
- **Optimization & Control:** Convex Optimization, Stochastic Optimization, Lyapunov Optimization
- **Mathematics:** Probability, Stochastic Processes, Game Theory, Queueing Theory
- **Deep Learning:** PyTorch, Transformers, Vision-Language Models
- **Tools:** Git, LaTeX

SERVICE & LEADERSHIP

- President, Mathematics Society, University of Moratuwa (2018–2019); organized M-Talks (monthly research seminars).
- Problem setter and organizer for national mathematics and programming competitions (uMora, MoraX-treme); designed problems and coordinated events.
- Conducted workshops in robotics, algorithmic programming, and problem-solving for university and school students.
- Led outreach initiatives through community programs (Grama Prabodhaya, Soyuru Sathkaraya), teaching mathematics and English in rural schools.

- Invited speaker, IEEE Student Branch University of Jaffna (2024); delivered session on problem-solving techniques.
- Volunteer for the 8th Learning for Dynamics & Control Conference (L4DC), 2026, and the 3rd International Conference on Neuro-Symbolic Systems (NeuS), 2026 Conference.